

Analysis of credit bureau data and mortgage fit statistics for FICO Score 10T and VantageScore 4.0

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¹ This paper reflects independent analysis performed by Milliman as part of a consulting engagement with FICO. The analysis reflects Milliman's own methodology and conclusions.

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Introduction

Since the late 1990s, Classic FICO has been relied upon by Freddie Mac and Fannie Mae (the Enterprises) for loan delivery. In 2020, the Federal Housing Finance Agency (FHFA) started a process to modernize the credit score models used for loan delivery to the Enterprises. As of May 2026, the FHFA has approved two credit scores, in addition to Classic FICO, for Enterprise loan delivery: VantageScore 4.0 and FICO Score 10T (FICO 10T). On April 22, 2026, the FHFA announced "...they [the Enterprises] will permit approved lenders to deliver mortgage loans using a credit score generated by either the Classic FICO model or the VantageScore 4.0 model."² The FHFA also announced that "[s]imilar implementation efforts are underway with respect to FICO 10T. The Enterprises expect to be able to publish historical FICO 10T scores in Summer 2026 and adopt scores from the model at a later date."³ This modernization represents a shift in the way mortgages will be underwritten and priced.

FICO 10T and VantageScore 4.0 both incorporate trended credit data—evaluating borrower behavior over a 24-month window rather than a point-in-time snapshot—a fundamental methodological departure from Classic FICO, which has been the industry standard since the late 1990s. While the Enterprises have published comparative performance data for VantageScore 4.0 and Classic FICO, data for FICO 10T has been provided to the FHFA and the GSEs but, as noted above, not publicly released at the time of this publication.

Milliman obtained a representative data sample directly from one of the three major U.S. credit bureaus, covering nearly 20 million new mortgage trades across multiple historical periods, with VantageScore 4.0 and FICO 10T available across all mortgage types. The data received from the credit bureau included performance flags to identify trades that had a default event or loan modification during the observation window. Milliman evaluated each score using various model fit statistics to assess their relative effectiveness in predicting mortgage default.

This paper, commissioned by FICO, evaluates the strength of the fit statistics for FICO 10T and VantageScore 4.0, the distribution across credit score bins, and the degree of loan-level consistency between the scores. This research helps market participants independently evaluate each credit score for underwriting and pricing mortgages.

Data

Milliman obtained six cross-sectional snapshots of credit bureau data at the individual trade level. The dataset contained various types of tradelines⁴ and was filtered to only mortgages that were new originations (originated up to six months prior to the archive date) for this analysis. Figure 1 summarizes key statistics of the data when filtered to new mortgage originations.

2. Federal Housing Finance Agency. (April 22, 2026). Credit scores. Retrieved April 28, 2026, from <https://www.fhfa.gov/policy/credit-scores>.

3. Ibid.

4. Tradeline is defined as a record of a credit account, including auto loans, student loans, personal loans, mortgages, etc.

FIGURE 1: CREDIT BUREAU DATA SUMMARY FOR NEW MORTGAGE ORIGINATIONS

ARCHIVE DATE	PERFORMANCE WINDOW		NUMBER OF MORTGAGES	DEFAULT RATE DURING THE PERFORMANCE WINDOW
	START	END		
2/1/2012	8/1/2011	8/1/2013	2,364,791	0.41%
4/1/2013	10/1/2012	10/1/2014	4,227,480	0.28%
4/1/2016	10/1/2015	10/1/2017	3,181,776	0.43%
4/1/2018	10/1/2017	10/1/2019	3,325,070	0.93%
4/1/2020	10/1/2019	10/1/2021	4,935,013	1.19%
4/1/2023	10/1/2022	10/1/2024	1,783,368	3.10%
Total			19,817,498	0.91%

The archive date represents the last month that would constitute a borrower as a new origination. The performance window start and end represent the period in which borrower performance is observed. Therefore, within each snapshot, 18–24 months of performance are observed for each borrower, with the number of performance months depending on the origination date. For each archive, credit scores were calculated prior to mortgage origination to best represent the credit score that would have been produced during the origination process.

For each trade record, a default is defined as a loan that was reported as a 90-day delinquency or worse (including foreclosure) during the performance window or a loan that received a loan modification prior to reaching 90 days delinquent. The reason for including loan modifications in the definition is that mortgage modifications are generally not provided to borrowers unless the borrower can demonstrate a financial hardship and has either missed a mortgage payment or will miss a mortgage payment.^{5,6} The exception to this requirement was during the COVID-19 pandemic, when documentation of financial hardship was not required and delinquency rates on mortgages rapidly increased and many borrowers entered voluntary forbearance programs. Furthermore, loan modifications result in additional costs to mortgage guarantors, servicers, and credit risk transfer (CRT) investors, and they impact prepayment speeds on mortgage-backed security (MBS) pools.

For this study, model fit statistics are aggregated across five archive periods: 2012, 2013, 2016, 2018, and 2023. The 2020 archive period is removed from this study due to the unique default rate experience resulting from the COVID-19 pandemic and resulting forbearance policies. The appendix provides additional details on the 2020 archive period.

This analysis segments mortgages into the following mortgage types:

1. All mortgages in the dataset
2. Loans delivered to Freddie Mac or Fannie Mae (Freddie/Fannie)
3. Loans endorsed by the Federal Housing Administration (FHA)

Freddie / Fannie and FHA loans are analyzed separately in this study because “In a joint announcement, HUD Secretary Scott Turner and FHFA Director William J. Pulte announced that the Federal Housing Administration and

5. Fannie Mae. (April 8, 2026). Servicing guide. Retrieved April 28, 2026, from <https://servicing-guide.fanniemae.com/svc/d2-3.2-06/fannie-mae-flex-modification>.

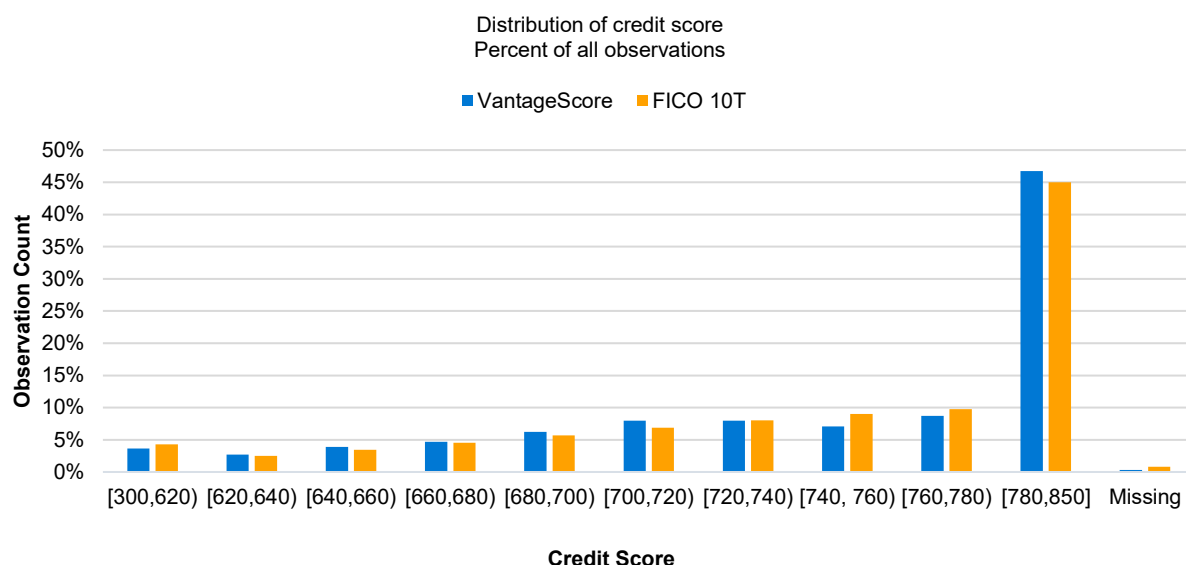
6. Federal Housing Administration. (n.d.). FHA's loss mitigation program. U.S. Department of Housing and Urban Development. Retrieved April 28, 2026, from <https://www.hud.gov/helping-americans/fha-loss-mitigation>.

Fannie Mae and Freddie Mac are implementing their first new credit score models for mortgages in decades.⁷⁴ referring to VantageScore 4.0 and FICO 10T.

Distribution of scores and measures of predictive power

Figure 2 provides a distribution of all mortgages by credit score for FICO 10T and VantageScore 4.0. Note, the distribution of mortgages will vary by mortgage type. For example, FHA loans have lower average credit scores relative to Freddie/Fannie loans. This chart provides the distribution for all mortgages. The credit score bins in the chart are the credit score bins utilized for pricing conventional mortgages through loan-level price adjustments. A bin is included for those mortgages without a valid credit score ("Missing"), though this is rare (for VantageScore 4.0, 0.3% of mortgages and for FICO 10T, 0.8% of mortgages).

FIGURE 2: DISTRIBUTION OF MORTGAGES BY CREDIT SCORE, ALL ARCHIVES EXCLUDING APRIL 2020



For this analysis, predictive power measures how well a credit score differentiates borrowers who are more likely to default from borrowers who are less likely to default. Three statistical measures are calculated to assess predictive power: the Kolmogorov-Smirnov statistic (KS-statistic), bottom decile lift, and area under the curve (AUC); the appendix of this paper provides a description of each measure. A summary of each statistical measure is provided here:

- KS-statistic measures how strongly the score distribution separates high-risk borrowers from low-risk borrowers.
- Bottom decile lift measures the percentage of total defaults identified in the lowest decile of the score distribution relative to a randomly assigned score.
- AUC measures the probability that a randomly selected default has a lower score compared to a randomly selected non-default borrower.

For all the calculated model fit statistics, a higher value indicates a stronger performing score.

When calculating model fit statistics, Milliman filtered the data to only include borrowers with values for all credit scores (i.e., the small portion of borrowers missing one or more credit scores were excluded from the analysis). This ensures an apples-to-apples comparison of model fit by using the same borrowers in the calculation of model fit statistics for each score. Figure 3 provides the number of mortgages and default rate by mortgage type. The default

⁷ Federal Housing Finance Agency. (April 22, 2026). Homebuying Advances into New Era of Credit Score Competition. Retrieved April 28, 2026, from <https://www.fhfa.gov/news/news-release/homebuying-advances-into-new-era-of-credit-score-competition>

rate is highest for FHA loans and lowest for Freddie/Fannie loans. Loans included in all mortgages that are not Freddie/Fannie or FHA loans include VA (Department of Veterans Affairs) and non-agency mortgages.

FIGURE 3: NUMBER OF MORTGAGES AND DEFAULT RATE BY MORTGAGE TYPE

MORTGAGE TYPE	NUMBER OF MORTGAGES	DEFAULT RATE
All Mortgages	14,882,817	0.82%
Freddie/Fannie	7,465,314	0.28%
FHA	2,226,900	3.01%

Figures 4-6 provide a summary of the statistical results. The results are shown for all mortgages and by mortgage type.

FIGURE 4: KS-STATISTIC, ALL ARCHIVES EXCLUDING APRIL 2020

MORTGAGE TYPE	NUMBER OF MORTGAGES	DEFAULT RATE	VANTAGESCORE 4.0	FICO 10T	DIFFERENCE	% DIFFERENCE
	A	B	C	D	E = D - C	F = E / C
All Mortgages	14,882,817	0.82%	57.9%	59.7%	1.8%	3.1%
Freddie/Fannie	7,465,314	0.28%	49.4%	51.1%	1.7%	3.4%
FHA	2,226,900	3.01%	36.0%	38.7%	2.7%	7.7%

The KS-statistic was 57.9% for VantageScore 4.0 and 59.7% for FICO 10T for all mortgages. Across all mortgages and by each mortgage type, FICO 10T has a higher KS-statistic relative to VantageScore 4.0. The largest difference between the KS-statistic is observed in FHA mortgages, where the percentage difference is 7.7% higher for FICO 10T relative to VantageScore 4.0.

FIGURE 5: BOTTOM DECILE LIFT, ALL ARCHIVES EXCLUDING APRIL 2020

MORTGAGE TYPE	NUMBER OF MORTGAGES	DEFAULT RATE	VANTAGESCORE 4.0	FICO 10T	DIFFERENCE	% DIFFERENCE
	A	B	C	D	E = D - C	F = E / C
All Mortgages	14,882,817	0.82%	5.72	6.04	0.32	5.7%
Freddie/Fannie	7,465,314	0.28%	4.81	5.06	0.25	5.3%
FHA	2,226,900	3.01%	3.14	3.39	0.25	8.2%

The bottom decile lift is an important metric for evaluating the ability of the credit score to identify borrowers with the highest likelihood of default. The higher the value, the more defaults are identified within the lowest 10% of credit scores in the data. This statistic is useful when evaluating underwriting criteria to determine whether the borrower will be successful in managing the debt. The bottom decile lift for FICO 10T is 6.04 across all mortgages, and the bottom decile lift for VantageScore 4.0 is 5.72 across all mortgages. The percentage improvement in the bottom decile lift is 5.7% for FICO 10T relative to VantageScore 4.0. Similar results are observed for Freddie/Fannie mortgages. The largest relative difference between scores is observed for FHA mortgages (+8.2% for FICO 10T), consistent with the KS-statistic.

FIGURE 6: AUC, ALL ARCHIVES EXCLUDING APRIL 2020

	NUMBER OF MORTGAGES	DEFAULT RATE	VANTAGESCORE 4.0	FICO 10T	DIFFERENCE	% DIFFERENCE
MORTGAGE TYPE	A	B	C	D	E = D - C	F = E / C
All Mortgages	14,882,817	0.82%	86.5%	87.3%	0.8%	0.9%
Freddie/Fannie	7,465,314	0.28%	81.9%	82.6%	0.7%	0.8%
FHA	2,226,900	3.01%	74.4%	76.0%	1.6%	2.2%

The AUC demonstrates the average ordinal ability of FICO 10T and VantageScore 4.0 to rank default risk across the entire credit score distribution, in contrast to the KS-statistic and the bottom decile lift, which focus on specific points of the distribution. Consistent with the KS-statistic and bottom decile lift, the AUC for FICO 10T is greater than VantageScore 4.0 across all mortgage types, with the greatest difference being observed for FHA mortgages.

Across the fit statistics measured, both scores display an ability to segment mortgage credit risk between high-risk and low-risk borrowers within the periods observed. The underlying data to calculate each credit score is pulled from the same sources of information on borrower credit profiles stored at the three major U.S. credit bureaus, but each score is built under a different formula.

Mortgages are priced across the credit score range of 300 to 850, and being in a different loan-level price adjustment (LLPA) band will typically result in different mortgage rates and prices. Given this, Milliman sought to quantify the degree of loan-level consistency between FICO 10T and VantageScore 4.0. Figure 7 shows the mortgage count in the data within cross sections of FICO 10T and VantageScore 4.0. The credit score bins are defined to be consistent with the current LLPAs from Freddie Mac and Fannie Mae.⁸

FIGURE 7: MORTGAGE COUNT BY FICO 10T AND VANTAGESCORE 4.0, ALL ARCHIVES EXCLUDING APRIL 2020

FICO 10T	Vantage										Total
	[300,620)	[620,640)	[640,660)	[660,680)	[680,700)	[700,720)	[720,740)	[740,760)	[760,780)	[780,850]	
[300,620)	345,878	135,944	95,135	43,780	17,043	6,433	1,826	609	209	88	646,945
[620,640)	70,640	81,224	97,435	68,806	35,610	15,480	4,577	1,293	508	210	375,783
[640,660)	48,534	75,320	122,972	118,050	84,430	46,147	15,470	4,279	1,556	783	517,541
[660,680)	25,833	54,214	119,248	153,947	153,384	107,349	45,437	13,790	5,107	2,790	681,099
[680,700)	11,309	28,021	83,307	146,227	211,251	197,715	107,003	40,209	16,235	9,658	850,935
[700,720)	4,818	11,592	41,153	97,643	207,313	276,743	206,783	102,215	50,804	31,540	1,030,604
[720,740)	1,226	3,092	14,083	44,248	132,393	266,057	293,841	205,552	139,770	103,666	1,203,928
[740,760)	418	985	3,960	14,428	58,555	167,654	268,631	271,548	265,097	300,069	1,351,345
[760,780)	90	249	974	4,294	20,884	75,028	159,257	227,749	319,609	659,120	1,467,254
[780,850]	37	85	336	1,570	9,648	40,017	93,420	192,862	511,845	5,907,563	6,757,383
Total	508,783	390,726	578,603	692,993	930,511	1,198,623	1,196,245	1,060,106	1,310,740	7,015,487	14,882,817

Despite any correlation between these two scores, only 53.5% of the loans are within the same credit score bin for LLPA pricing. In addition to influencing mortgage pricing and mortgage rates, default rates between adjacent credit score bins increase by a factor between 1.5 and 3.0 in the data. Figure 8 calculates the default rates for the same grid. Note, for credibility, bins in Figure 8 with fewer than 30 defaults are labeled as NA.

8. Note, the April 22 announcement from the FHFA indicated different LLPA grids are being developed for VantageScore 4.0 (and presumably FICO 10T). Since those grids have not been published, the existing LLPA grids are used for this analysis.

FIGURE 8: DEFAULT RATE BY FICO 10T AND VANTAGESCORE 4.0, ALL ARCHIVES EXCLUDING APRIL 2020

	Vantage										
FICO 10T	[300,620)	[620,640)	[640,660)	[660,680)	[680,700)	[700,720)	[720,740)	[740, 760)	[760,780)	[780,850]	Total
[300,620)	9.08%	5.84%	5.03%	4.66%	4.64%	4.59%	2.85%	NA	NA	NA	7.31%
[620,640)	4.75%	3.93%	3.47%	3.16%	2.93%	2.56%	2.05%	NA	NA	NA	3.64%
[640,660)	3.70%	3.14%	2.74%	2.43%	2.28%	1.79%	1.65%	1.61%	NA	NA	2.61%
[660,680)	2.85%	2.42%	2.09%	1.84%	1.63%	1.43%	1.19%	0.92%	0.84%	NA	1.78%
[680,700)	2.41%	1.86%	1.54%	1.37%	1.18%	1.03%	0.88%	0.73%	0.66%	0.43%	1.18%
[700,720)	1.87%	1.43%	1.19%	1.06%	0.90%	0.74%	0.63%	0.55%	0.44%	0.32%	0.76%
[720,740)	NA	1.16%	0.91%	0.82%	0.66%	0.54%	0.48%	0.40%	0.35%	0.23%	0.48%
[740, 760)	NA	NA	0.78%	0.62%	0.50%	0.41%	0.33%	0.30%	0.24%	0.17%	0.29%
[760,780)	NA	NA	NA	NA	0.36%	0.29%	0.23%	0.23%	0.21%	0.12%	0.18%
[780,850]	NA	NA	NA	NA	0.45%	0.24%	0.18%	0.14%	0.13%	0.05%	0.07%
Total	3.50%	2.33%	1.75%	1.08%	0.66%	0.48%	0.38%	0.20%	0.06%	0.00%	0.82%

Holding one score constant and observing default rates across the opposite score, there is a consistent decreasing relationship between the secondary score and the default rate. For example, when looking at the Vantage [780, 850] column, borrowers who have a FICO 10T of [680, 700) display a higher default rate (0.43%) relative to borrowers who have a FICO 10T of [780, 850] (0.05%). This relationship applies in the other direction as well—when FICO 10T is within [780, 850], the default rate for borrowers who also have a [680, 700) VantageScore 4.0 is higher (0.45%) than borrowers who also have a VantageScore 4.0 within [780, 850] (0.05%).

This observation demonstrates that, given the degree of variation in score bin assignment that exists between the two scores, use of both scores can provide added information about a single borrower. This variation in score bin assignment also forms the basis of concerns of adverse selection in a lender choice setting, as covered in Milliman's prior research⁹. There is complexity in developing a rules-based system for approval and pricing using multiple credit scores, particularly when the credit scores may provide a different risk profile of the borrower. In addition, the Q&A from the FHFA's April announcement stated, "Approved lenders will have the choice to report credit scores from either Classic FICO or VantageScore 4.0. For the time being, the Enterprises will not accept scores from multiple models on a given loan at the time of loan delivery."¹⁰ Therefore, it is unlikely that the information gained from using multiple credit scores will be used for loan pricing and delivery for Freddie Mac and Fannie Mae mortgages.

Summary of analysis

This analysis presented three measures of predictive power for FICO 10T and VantageScore 4.0 for evaluating mortgage default risk. Across all measures and mortgage types, we find that FICO 10T consistently produced the highest measure of predictive power across all fit statistics and mortgage types.

Given the credit scoring models use different algorithms to evaluate creditworthiness, there will be pricing differences under the current LLPA grids between the two scores. This is highlighted by the fact that only 53.5% of the consumers in the data would have the same LLPA pricing (assuming the LLPA grids are the same for VantageScore 4.0 and FICO 10T) if all mortgages were conventional mortgages.

For deliveries to Freddie Mac and Fannie Mae, information on how LLPA's will be adjusted for VantageScore 4.0 or FICO 10T has not been released, and there are many questions on how mortgages will be priced under multiple credit scores. These questions include: How are the LLPA grids going to be calibrated to not penalize or favor a given credit score? How will the LLPA grids account for the bias of lender choice? The current policy, which was re-iterated in the April 22 Q&A section, is to allow lenders to choose one score (on an individual loan-by-loan basis) to deliver the loan. Prior research from Milliman has demonstrated this policy will bias default rates¹¹ without offsetting controls.

⁹ Glowacki, J., Huff, R., & Ludden, B. (October 25, 2025). "Lender choice" introduces a bias to default rates for mortgage underwriting. Milliman. Retrieved April 28, 2026, from <https://www.milliman.com/en/insight/lender-choice-introduces-bias-mortgage-underwriting>.

¹⁰ Federal Housing Finance Agency. (April 22, 2026), op. cit.

¹¹ Glowacki, J., Huff, R., & Ludden, B. (October 25, 2025). "Lender choice" introduces a bias to default rates for mortgage underwriting. Milliman. Retrieved April 28, 2026, from <https://www.milliman.com/en/insight/lender-choice-introduces-bias-mortgage-underwriting>.

If the policy of lender choice is adopted by Freddie Mac and Fannie Mae, will a similar policy be adopted by other participants in the mortgage market? Each of these questions remains outstanding.

Milliman remains active with the industry and this topic, and we are supporting our clients by:

- Providing research on the relative performance and predictive power of each credit score model. Additional research will provide a deeper dive into relative score performance between mortgage types (e.g., FHA, Freddie/Fannie, etc.) and segmenting the data into first-time home buyers and repeat mortgages.
- Providing analysis to update underwriting criteria and overlays to manage risk under multiple credit scores or a single credit score under lender choice
- Calibrating and validating mortgage performance models to use multiple credit scores or a single credit score under lender choice
- Calibrating mortgage pricing, servicing valuations, and credit guarantee fees to adjust for multiple credit scores or a single credit score under lender choice

Limitations

There are limitations to the analysis presented in this report. This section describes the known limitations of the analysis.

Borrowers in this sample were approved for a mortgage under Classic FICO. As a result, the dataset is censored to only borrowers with a Classic FICO score that met the underwriting guidelines, given the other risk attributes of the borrower. This censoring effect means that model fit statistics will likely be understated relative to the true predictive power of each score without the presence of this censoring. The results of the study do not attempt to make a judgment on the direction or magnitude of this effect on any credit score.

This analysis used one specific default definition (either 90-plus-day delinquency or a loan modification occurred within the performance window). Milliman also performed the analysis without including loan modification in the default definition, and the results of that analysis were consistent with the findings presented in this paper.

The scores obtained for this report were obtained from one credit bureau. This provides consistency in the scores between borrowers, but it is not consistent with the decision score used for underwriting and pricing mortgages. When mortgages are underwritten, credit scores are pulled from all three credit bureaus and aggregated using the tri-merge approach.

Mortgage default rates occur over the life of the mortgage. This analysis is limited to an 18-to-24-month performance window. This performance window is consistent with the default horizon used to develop credit scoring models (which are estimated on all types of credits) but less than the life of a mortgage. Milliman performed an analysis on 24-month default rates relative to lifetime default rates, and we found there is a high correlation (above 90%) between 24-month default rates and lifetime default rates.

Mortgage defaults are influenced by both the credit profile of the borrower and economic conditions. The time periods used in this analysis generally represent a period of strong mortgage credit performance and home price appreciation. Mortgage defaults have historically been elevated during periods of home price declines. The data available and required to obtain a VantageScore 4.0 or FICO 10T are limited to approximately calendar year 2012 and subsequently, and the data required to evaluate the relative performance of these credit scores during periods where home prices declined do not exist.

The analysis focuses on five specific observation periods. While the results demonstrated higher fit statistics for FICO 10T over VantageScore 4.0 within these periods across loan segments, a more granular analysis or an analysis conducted over different time periods may find instances where the inverse is observed.

The presentation of fit statistics provides insight into what model provided the best statistical power across the sample period. The fit statistics are all higher for FICO 10T, but this analysis does not attempt to evaluate the statistical significance or economic impact of these differences across segments. Further analysis is required, and

encouraged, to evaluate the impact of credit score modernization on changes to mortgage approval and denial rates, pricing, and costs across the entire mortgage lifecycle.

Appendix

MEASURES OF PREDICTIVE POWER

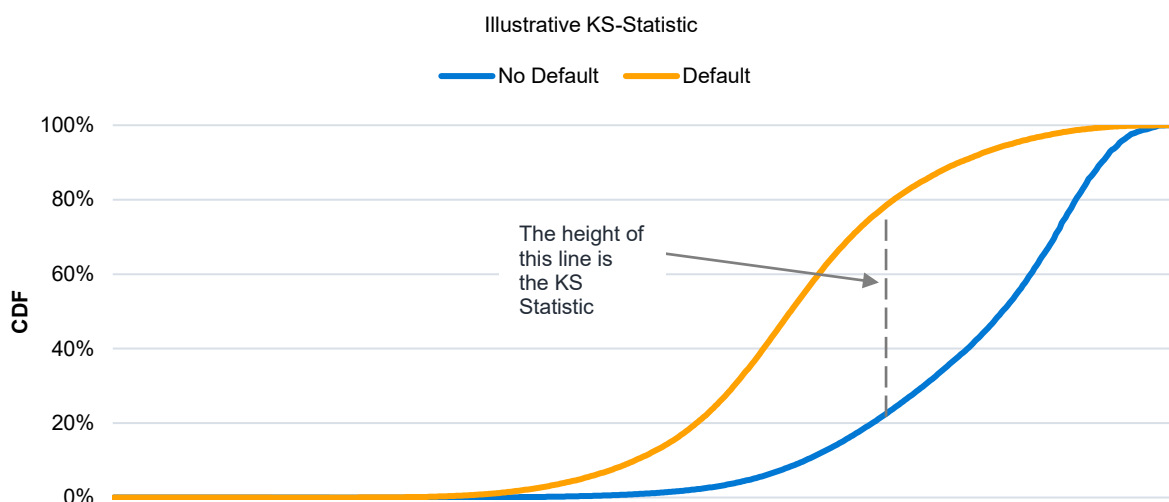
For this analysis, predictive power measures how well a credit score differentiates borrowers who are more likely to default from borrowers who are less likely to default. Three statistical measures were calculated to assess predictive power.

KS-statistic

The Kolmogorov-Smirnov statistic (KS-statistic) is taken from the K-S Test, which measures how similar the distribution of credit score is for borrowers who default and borrowers who did not default. Specifically, the KS-statistic is the maximum vertical distance between the cumulative distribution functions (CDF) of credit score for borrowers with and without a default. Higher values indicate that the credit scores have a stronger ability to separate defaulters from non-defaulters. A random score would have a KS-statistic of approximately zero.

For demonstration purposes, Figure 9 provides an illustration of the calculation of the KS-statistic.

FIGURE 9: ILLUSTRATIVE KS-STATISTIC CALCULATION



Bottom decile lift

The bottom decile lift represents the capture of defaulting borrowers in the lowest score decile relative to a random score. To calculate the bottom decile lift, each score is broken into 10 evenly sized deciles. Then, the number of defaults in the bottom decile is divided by the total number of defaults in the data. A random score would identify 10% of all defaults in the bottom decile. Therefore, the lift is calculated as this percentage of defaults identified divided by 10%. Higher lift indicates a stronger capability to segment a greater proportion of defaulting borrowers into the lowest score echelon. A random score would have a bottom decile lift of 1.

For demonstration purposes, Figure 10 shows the calculation performed to determine the bottom decile lift.

FIGURE 10: CALCULATION OF BOTTOM DECILE LIFT

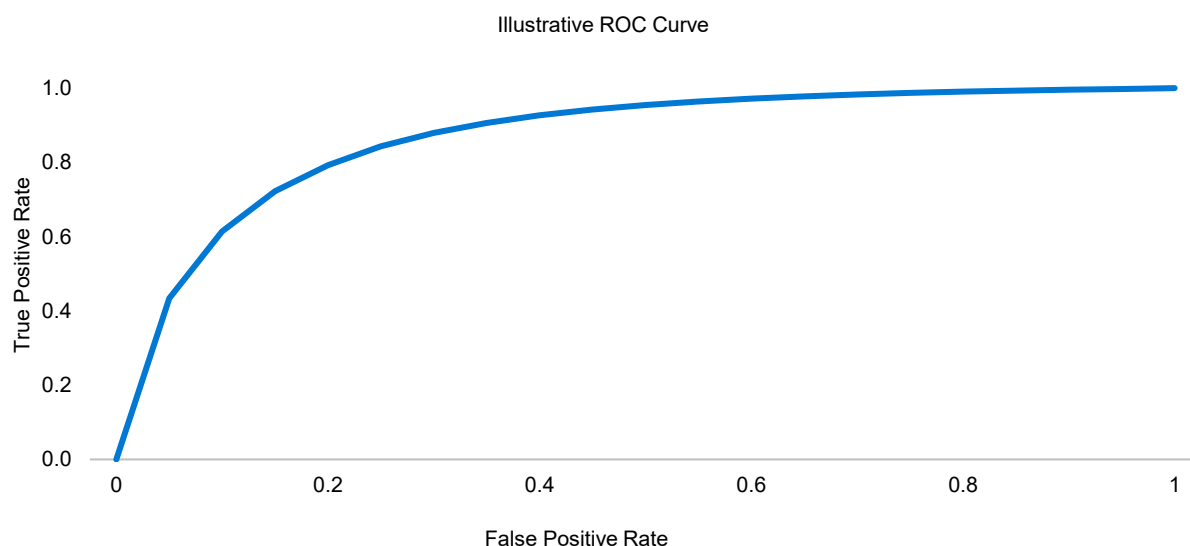
DECILE	NUMBER OF MORTGAGES	NUMBER OF DEFAULTS	DEFAULT RATE	LIFT
1	1,488,290	65,882	4.43%	5.43
2	1,488,287	24,951	1.68%	3.74
3	1,488,280	12,157	0.82%	2.83
4	1,488,280	7,144	0.48%	2.27
5	1,488,280	4,311	0.29%	1.89
6	1,488,280	2,516	0.17%	1.61
7	1,488,280	1,627	0.11%	1.40
8	1,488,280	1,161	0.08%	1.23
9	1,488,280	913	0.06%	1.10
10	1,488,280	740	0.05%	1.00
Total	14,882,817	121,402	0.82%	

The bottom decile lift is calculated as 65,882 divided by 12,140 (which is 10% of the total number of defaults).

Area under the curve

The area under the receiving operating characteristic curve (AUC) is calculated as the probability that a borrower who defaults was assigned a lower score than a borrower who did not default. For example, an AUC of 80% means that if a defaulting borrower and a non-defaulting borrower are selected at random, there is an 80% chance the defaulting borrower has a lower score than the non-defaulting borrower. A higher AUC indicates the model has stronger ranking capabilities. A randomly assigned score would have an AUC of 50%. AUC is used to evaluate how well a model differentiates risks across the entire range of scoring outcomes.

For demonstration purposes, Figure 11 provides a sample of the receiver operating characteristic (ROC) curve for illustrative purposes.

FIGURE 11: ILLUSTRATIVE EXAMPLE FOR ROC CURVE

2020 archive period

For this analysis, Milliman excluded loans originated in the 2020 archive period from the study. 2020 originations were impacted by the COVID-19 pandemic, and the default rates observed in the credit bureau data were inconsistent with default rates from publicly available mortgage performance data filtered to the same origination period and observation window.

This inconsistency is attributable to mortgage forbearance programs and credit reporting changes due to the Coronavirus Aid, Relief, and Economic Security (CARES) Act.¹² According to a blog post by the Consumer Financial Protection Bureau titled “Protecting your credit during the coronavirus pandemic,” lenders could provide accommodations or not provide accommodations to borrowers impacted by the COVID-19 pandemic.¹³ For federally backed mortgages (e.g., Freddie Mac, Fannie Mae, and FHA, VA, and U.S. Department of Agriculture mortgages), forbearance (i.e., accommodation) must have been provided to any borrower who claimed financial hardship due to the COVID-19 pandemic. For borrowers who received accommodation:

“How your lenders report your account to credit reporting agencies under the CARES Act depends on whether you are current or already delinquent when this agreement is made. These reporting requirements apply **only** if you are making any payments required by the agreement.

- If your **account is current** and you make an agreement to make a partial payment, skip a payment, or other accommodation, then the creditor is to report to credit reporting companies that you are **current on your loan or account**.
- If your account is already delinquent and you make an agreement, then the creditor cannot report you as **more delinquent** (such as reporting you as 60 days delinquent when you started out 30 days delinquent) during the period of the agreement.
- If your account is already delinquent and you make an agreement, and you bring your account **current**, the creditor must report that you are current on your loan or account.

This CARES Act requirement applies only to agreements made between January 31, 2020 and the later of either:

- 120 days after March 27, 2020 or
- 120 days after the national emergency concerning COVID–19 ends.”¹⁴

The national emergency concerning COVID-19 ended on May 11, 2023.¹⁵ Therefore, the reporting for delinquent loans during this period is impacted by the CARES Act and forbearance options for most mortgages in the data.

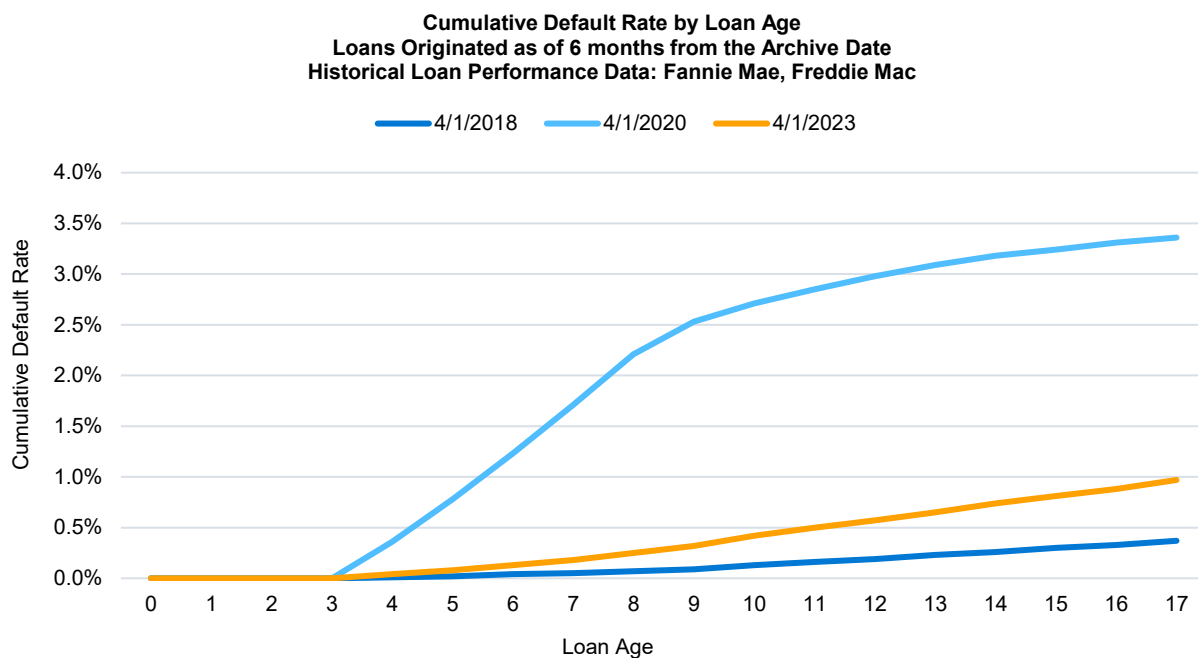
Milliman compared the default rates from the credit bureau data relative to the default rates in loan performance data published by Freddie Mac and Fannie Mae. Figure 12 shows cumulative development curves for 90-plus-day delinquencies from data published by Freddie Mac and Fannie Mae for the same loan population for the 2018, 2020, and 2023 archive periods.

12. Fiano, L. (July 29, 2020). Protecting your credit during the coronavirus pandemic. Consumer Financial Protection Bureau. Retrieved April 28, 2026, from <https://www.consumerfinance.gov/about-us/blog/protecting-your-credit-during-coronavirus-pandemic/>.

13. Ibid.

14. Ibid.

15. Centers for Disease Control and Prevention. (September 12, 2023). End of the federal COVID-19 public health emergency (PHE) declaration. Retrieved April 28, 2026, from https://archive.cdc.gov/www_cdc_gov/coronavirus/2019-ncov/your-health/end-of-phe.html.

FIGURE 12: CUMULATIVE 90-PLUS-DAY DELINQUENCY RATE BY LOAN AGE

From Figure 12, defaults from the COVID-19 period originations were significantly greater than defaults from 2018 and 2023 origination periods. For the 2020 origination period, cumulative 90-plus-day default rates approached 3.5%, relative to less than 1.0% for 2018 and 2023 originations. The data reported by Freddie Mac and Fannie Mae identify loans as being in forbearance, but they also report on the number of total missed payments. This reporting is different from the credit bureau data because it reflects actual mortgage payments made, which impact cash flows to MBS and CRT investors.

Figure 13 provides the cumulative default rates observed in the credit bureau data for Freddie Mac and Fannie Mae mortgages for the same reporting periods and exposure window.

FIGURE 13: TRADE PERFORMANCE DATA, FREDDIE/FANNIE MORTGAGES ONLY

Credit Bureau Trade Performance Data, Freddie/Fannie Mortgages Only		
ARCHIVE DATE	NUMBER OF MORTGAGES	DEFAULT RATE
4/1/2018	1,606,342	0.37%
4/1/2020	2,430,851	0.53%
4/1/2023	756,029	1.09%

The default rate for 2020 originations is materially different from the default rate observed in the Freddie Mac and Fannie Mae loan performance data. It is not clear from the data which mortgages are reported as a default and which mortgages are not reported as a default for this period. Given the low default rate in the credit bureau data and unique reporting circumstances with respect to the COVID-19 pandemic, this archive period was excluded from this study.

For completeness, Figure 14 provides the three fit statistics by archive period Fannie/Freddie mortgages. For the 4/1/2020 archive, the fit statistics are lower relative to all other archives, showing the influence of the COVID-19 pandemic on the model fit statistics.

FIGURE 14: MODEL FIT STATISTICS BY ARCHIVE PERIOD, NEW ORIGINATIONS

ARCHIVE	NUMBER OF MORTGAGES A	DEFAULT RATE B	VANTAGESCORE 4.0 C	FICO 10T D	DIFFERENCE E = D - C	% DIFFERENCE F = E / C
KS-statistic						
2/1/2012	1,347,303	0.12%	52.4%	54.1%	1.7%	3.3%
4/1/2013	2,250,879	0.11%	53.0%	55.0%	2.0%	3.8%
4/1/2016	1,504,761	0.16%	48.7%	50.0%	1.3%	2.7%
4/1/2018	1,606,342	0.37%	48.3%	49.9%	1.6%	3.4%
4/1/2020	2,430,851	0.53%	30.8%	32.1%	1.3%	4.4%
4/1/2023	756,029	1.09%	40.5%	43.4%	3.0%	7.4%
Bottom Decile Lift						
2/1/2012	1,347,303	0.12%	5.29	5.57	0.28	5.3%
4/1/2013	2,250,879	0.11%	5.57	5.71	0.14	2.5%
4/1/2016	1,504,761	0.16%	4.94	5.10	0.17	3.4%
4/1/2018	1,606,342	0.37%	4.63	4.75	0.12	2.7%
4/1/2020	2,430,851	0.53%	3.04	3.12	0.08	2.6%
4/1/2023	756,029	1.09%	3.74	4.04	0.30	8.1%
Area under the Curve						
2/1/2012	1,347,303	0.12%	83.2%	84.0%	0.8%	1.0%
4/1/2013	2,250,879	0.11%	82.9%	83.7%	0.8%	1.0%
4/1/2016	1,504,761	0.16%	81.0%	81.7%	0.8%	1.0%
4/1/2018	1,606,342	0.37%	81.4%	82.0%	0.6%	0.8%
4/1/2020	2,430,851	0.53%	70.7%	71.5%	0.8%	1.2%
4/1/2023	756,029	1.09%	76.6%	78.1%	1.5%	1.9%

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